higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE (VOCATIONAL)

PLANT AND EQUIPMENT
NQF LEVEL 3

(12030263)

19 November 2019 (X-Paper)
09:00–12:00

This question paper consists of 7 pages.

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TIME: 3 HOURS
MARKS: 100

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Start each section on a NEW page.
5. Use only a pen with blue or black ink.
6. Write neatly and legibly.

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QUESTION 1

Choose a description from COLUMN B that matches a term in COLUMN A. Write only the relevant letter (A–H) next to the question number (1.1–1.5) in the ANSWER BOOK.

COLUMN A		COLUMN B	
1.1	Class A	A	ordinary materials, such as wood, paper, coal, grass and textiles
1.2	Class B	B	electrical installations, such as cables, transformers and generators
1.3	Class C	C	flammable liquids, such as oil and petrol
1.4	Class D	D	high-pressure flammable gases, such as methane and propane
1.5	Class E	E	metals that burn at high temperature, such as aluminium and sodium
		F	heat, oxygen and fuel
		G	low-burning metals
		H	non-combustible materials

(5 × 1) **[5]**

QUESTION 2

2.1 The following steps are guidelines to follow when using a chain lever hoist successfully.

REARRANGE the question numbers (2.1.1–2.1.8) in the ANSWER BOOK.

2.1.1 Attach the upper hook to the suspension point that can handle the weight.

2.1.2 Balance the load correctly before hoisting it.

2.1.3 Make sure that the load is properly fixed to the bottom hook.

2.1.4 Turn the selector to 'Lift/Up'. It has three positions: 'Lift/Up', 'Neutral' or 'Down'.

2.1.5 Check that the chain is in a good condition.

2.1.6 Make sure that the load is securely attached to the hook.

2.1.7 Check that the safety clips on both hooks are closed correctly.

2.1.8 Pump the lever up and down to raise and lower the load.

(8 × 1) (8)

2.2 Give definitions for each of the following terms:

2.2.1 Jack

2.2.2 Safe working load

2.2.3 Safety factor

(3 × 2) (6)

2.3 List SIX hand signals that a banksman can give to a crane operator during lifting situations. (6)

[20]

QUESTION 3

3.1 Complete the following paragraph by filling in the missing terms. Write the correct answer next to the question numbers (3.1.1–3.1.3) in the ANSWER BOOK.

(3.1.1) … is defined as a twisting force and (3.1.2) … is the rate at which work is done, while (3.1.3) … is a force acting over a distance. (3 × 1) (3)

3.2 State FOUR advantages of diesel engines over petrol engines. (4)

3.3 List TEN parts of an internal-combustion engine. (10)

3.4 State the percentage efficiency of the following engines:

3.4.1 Electric motors

3.4.2 Petrol engines

3.4.3 Diesel engines

(3 × 1) (3)
[20]

QUESTION 4

4.1 Discuss a torque converter under each of the following headings:

4.1.1 Definition of a torque converter (1 × 1) (1)

4.1.2 Where to use a torque converter (2 × 1) (2)

4.1.3 Purpose of a torque converter in equipment (1 × 1) (1)

4.1.4 TWO examples of equipment in which a torque converter is used. (2 × 1) (2)

4.2 State TWO limitations of a diesel–electrical drive. (2)

4.3 Describe TWO ways in which to treat your vehicle with care when changing gears. (2)
[10]

QUESTION 5

5.1 State FIVE basic steps to follow when checking the oil level of a vehicle. (5)

5.2 List SIX warning lights that may be displayed on the vehicle cluster/panel to warn the driver about vehicle conditions. (6)

5.3 Give the meaning of the following abbreviations that are used in vehicle loading:

5.3.1 GVM

5.3.2 GCM

(2 × 1) (2)

5.4 A multimeter is an instrument used to check the voltage of a battery, and some multimeters have display and LED lights which scan the voltage of the battery.

Indicate what the following colours refer to when the voltage of a battery is checked using a multimeter that has display and LED lights:

5.4.1 Green light

5.4.2 Yellow light

5.4.1 Red light

(3 × 1) (3)

5.5 List FOUR items that will need to be replaced when a minor service is performed on a vehicle. (4)

[20]

QUESTION 6

6.1 State FIVE tools that can be run by compressed air. (5)

6.2 Identify the requirements for a good circular saw. (5)

6.3 Name FIVE main parts of an angle grinder. (5)

[15]

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QUESTION 7

7.1 Differentiate between each of the following:

7.1.1 Acetylene cylinders

7.1.2 Oxygen cylinders

(2 × 3) (6)

7.2 Describe FOUR steps to be followed when shutting the system down after working with an oxy-acetylene set. (4)

[10]

TOTAL: 100

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